पृथ्वी विज्ञान मंत्रालय
MINISTRY OF
EARTH
SCIENCES
सत्यमेव जयते

G20 भारत 2023 INDIA

MINISTRY OF EARTH SCIENCES (MOES)

COMMITTED TO SCIENCE FOR PUBLIC SERVICE & NATION-BUILDING

MoES holistically addresses all five components of the earth system: atmosphere (air), hydrosphere (water), cryosphere (frozen solid), lithosphere (land) and biosphere (life) and their complex interactions. It is mandated to provide reliable services for weather, climate, ocean and coastal state, hydrology, seismology, and natural hazards; explore and harness marine living and non-living resources sustainably; and explore the two poles of the Earth (Arctic, Antarctic) Himalayas and Southern Ocean. MoES services are utilised by international and national agencies, state governments, local bodies, and academia for public and social benefits and welfare.

MoES activities are carried out under six umbrella schemes: Atmosphere and Climate Research–Modelling, Observing Systems and Services (ACROSS); Ocean Services, Modelling, Application, Resources and Technology (O-SMART); Polar and Cryosphere Research (PACER); Seismology and Geosciences (SAGE); Research, Education, Outreach and Training (REACHOUT); and Deep Ocean Mission (DOM).

MoES is headquartered in New Delhi and its activities and mandate is pursued through eleven institutes: Centre for Marine Living Resources and Ecology (CMLRE); National Centre of Coastal Research (NCCR); National Centre for Seismology (NCS); National Centre for Medium-Range Weather Forecasting (NCMRWF); India Meteorological Department (IMD); National Institute of Ocean Technology (NIOT); Indian National Centre for Ocean Information Services (INCOIS); National Centre for Polar and Ocean Research (NCPOR); Indian Institute of Tropical Meteorology (IITM); National Centre for Earth Science Studies (NCESS); and Borehole Geophysics Research Laboratory (BGRL). The website of MoES is www.moes.gov.in.

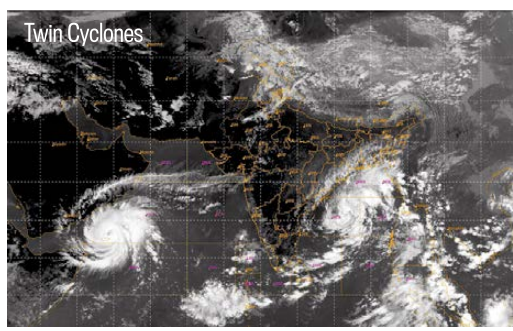
In 2021, Dr M Ravichandran took charge as the Secretary of MoES. Over the years, India has enhanced its prowess in weather, climate, ocean and seismological observation systems, research and development, and capabilities in forecasting, prediction, and dissemination of information to stakeholders and last-mile users. There are 39 Doppler Weather Radars for better observation and prediction of extreme weather events spread across the country, including in high-altitude regions and the northeast. About 200 agro-automated weather stations help make forecasts more accurate. There are 26 meteorological centres that are specialised for observing and disseminating information, advisories, and warnings about regional weather. India has 63 Pilot Balloon stations for winds-aloft observations (wind speeds and direction); 1200 nowcast stations; 1300 city forecast stations; 1008 Automated Weather Stations; 1382 Automated Rain Gauges, and quite a vast lightning location network of 83 sensors, which make our weather- and climate-related services amongst the best in the world. India's cyclone and tsunami early warning centres are recognised globally for providing services to 13 and 25 countries respectively. The agro-met and fishing zone advisories help the agricultural and fisherfolk community in information-backed planning and decision-making. Farmers use agro-met advisories for weather information-based management like sowing, irrigation, fertiliser and pesticide application, and harvest, and fisherfolk use fishing zone advisories to improve catch and fuel savings. Investments of approximately rupees one thousand crores in the Monsoon Mission and



> "Under the leadership of Hon'ble PM, Sh Narendra Modi ji, implementation of the Deep Ocean Mission of the Govt of India endeavours to harness the boundless potential of the ocean and fortify India's blue economy."
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> ***—Shri Kiren Rijiju, Hon'ble Union Minister***



Sagar Nidhi


Twin Cyclones